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Thermodynamic assessment of the Ga–La and Ga–Pr systems supported by ab-initio calculations

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ABSTRACT

Ab-initio calculations within density functional theory (DFT) were used to calculate the enthalpies of formation of the binary intermetallics in the Ga–La and Ga–Pr systems. The thermodynamic data and the phase diagram of these systems have been critically assessed according to the Calphad technique. The primary solid solutions in the two systems were modeled with Redlich–Kister polynomial, while the associate model was used to describe the short-range ordering behavior of the liquid phase.

The intermetallics compounds Ga_6La, Ga_4La, Ga_2La, GaLa, Ga_3La_5, $GaLa_3$, Ga_6Pr, Ga_2Pr, GaPr, Ga_3Pr_5 and $GaPr_2$ in addition to the suspected compound Ga_3Pr are taken into consideration in this optimization. The calculated enthalpies of formation using the VASP code are used in the optimization of the Ga–La and Ga–Pr phase diagrams which agrees well with the experimental data.

1. Introduction

Rare earth (RE) elements with gallium based multi-alloys are characterized by interesting magnetic behaviors and attract much attention. The intermetallics of these systems are used as magnetic refrigeration materials which are based on the use of the magneto-thermodynamic phenomenon known as the MagnetoCaloric Effect [1,2]. The magnetocaloric effect includes the temperature change of the magnetic material under the action of an external magnetic field. The challenge is to have high-magnetic refrigeration efficiency near room temperature at relatively low applied magnetic fields [3].

Therefore to understand the technological applications of these alloys, it is necessary to know well the exact phase diagrams and the thermodynamic properties of these systems. Before proceeding to the study of the ternary systems based on rare earth and gallium elements, RE-Ga binary systems must be assessed. In this work the enthalpies of formation of the intermetallic compounds in the Ga–La and Ga–Pr systems are determined using ab-initio calculations, the assessment of the all thermodynamic functions of the systems using the Calphad approach supported by our ab-initio results and the phase diagrams of the two systems will be established.

2. Literature data

2.1. Ga–La system

The Gallium–Lanthanum binary system has been investigated by different scientists. Works carried out up to 1991 have been reviewed by Palenzona and Cirafici [4]. The Ga–La phase diagram was firstly studied by Yatsenko [5] and Yatsenko et al. [6], using thermal analysis, microscopy, and X-ray diffraction method. These authors reported the existence of four intermediate phases: Ga_2La, GaLa, Ga_3La_5 and $GaLa_3$. The first compound melts congruently at 1450 °C. GaLa, Ga_3La_5 and $GaLa_3$ were formed by a peritectic reaction at 1010 °C, 825 °C and 615 °C respectively. There were also two eutectics reactions: Liq $\leftrightarrow Ga_2La + Ga$ at 30 °C and Liq$\leftrightarrow GaLa_3+\beta La$ at 550 °C and about 80 at.% La. To describe the eutectic on the Ga-rich side, Yatsenko [5] and Yatsenko et al. [6] used the term "degenerate" probably because the temperature of the solids does not change significantly from the melting temperature of Ga. No significant solid solubility of La in (Ga) or Ga in (La) was observed and all intermediate phases unless Ga_2La appeared to be linear compounds. Pelleg et al. [7] studied the Ga-rich side of Ga-RE systems and found the compound Ga_6La formed by a peritectic reaction at 477 °C. However Yatsenko [5] found an unknown thermal effect at 726 °C, while Lu at al [8]. fixed the decomposition temperature of Ga_6La at 700 °C. Dayan et al. [9], using interdiffusion studies of lanthanum

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with gallium determined the range of homogeneity of Ga_2La and found that it extends on the Ga-rich side from 18 to 33.3 to. %. Their study also excludes the possible existence of Ga_3La because no layer of this composition is present in this system. Lu at al [8]. studied the Ga-rich side of the Ga–La system. They reported that Ga_2La has an homogeneity range extending from 28.4 to 33.3 at.% La, which occurs when one La atom is replaced randomly by a pair of Ga atoms.

The Ga–La phase diagram evaluated by Palenzona and Cirafici [4] based on the work of Yatsenko [5] and Yatsenko et al. [6] is given in the compilation of Massalski [10].

Zevin et al. [11] obtained by X-ray diffraction examination that an orthorhombic Ga_4La compound exists at least in the Ga–La system. More recently, a new type of phase transformation was proposed by Kaplan and Kimmel [12,13]. The existence of the ordered Ga_4La; structure (ϵ´) has been experimentally verified. The Ga_4La (ϵ´) phase was described as a long-range ordered structure with a stoichiometric concentration of 80 at. % Ga. The ordering temperature is 480 °C.

Crystal structure and lattice parameters of the pure elements and six compounds (Ga_6La, Ga_4La, Ga_2La, GaLa, Ga_3La_5 and $GaLa_3$) are listed in Table 1 [7,12,14–19].

Vnuchkova et al. [20] determined the enthalpies of formation of the intermetallic compound Ga_2La by the emf method. On the basis of their calorimetric measurement at 1373 K, Meshel and Kleppa [21] measured the standard enthalpy of formation of the Ga_2La compound and found the value of −69.2 ± 2.4 kJ/mol. Babu et al. [22] determined the standard enthalpies of formation of Ga_2La, GaLa and Ga_3La_5 by high temperature liquid gallium solution calorimetry. De Boer et al. [23] calculated by a semi-empirical model the enthalpies of formation of the binary compounds Ga_2La, GaLa, Ga_3La_5 and $GaLa_3$. Recently Yang et al. [24] calculated the enthalpy of formation of the Ga_6La, Ga_2La, GaLa and Ga_3La_5 compounds by the Vienna ab-initio Simulation Package (VASP) code.

The phase diagram of the Ga–La system has been optimized thermodynamically by Idbenali and Servant [25] based essentially on the discrepancy thermodynamic data of the researchers cited above. The calculated liquidus is lower than the experimental data, especially in the Ga rich side assessed Ga–La, where there was a significant difference with the phase diagram evaluated by Palenzona and Cirafici [4]. For this reason the Ga–La system is reassessed based on the calculated enthalpies of formation of the intermetallic compounds and the experimental data from the literature cited above.

Table 1
The crystal structure data of the Ga–La and Ga–Pr systems.

phase	Pearson symbol and prototype	Space group	Lattice parameters (°A)			Ref.
			a	b	c	
Ga	oS8 (αGa)	Cmca	4.524	7.661	4.519	[14]
			4.573	7.7496	4.603	This work
αLa	hP4(αLa)	P63/mmc	3.765	–	12.150	[19]
			3.771	–	12.064	This work
αPr	hP4(αLa)	P63/mmc	3.672	–	11.834	[19]
			3.639	–	11.745	This work
Ga_6La	tP14 (Ga_6Pu)	P4/nbm	6.094	–	7.692	[7]
			6.098	–	7.785	This work
Ga_4La	oP20	Pmmm	4.403	7.542	13.015	[12]
			4.355	7.745	12.956	This work
Ga_2La	hP3(AlB_2)	P6/mmm	4.320	–	4.416	[15]
			4.336	–	4.452	This work
GaLa	oC8(CrB)	Cmcm	4.5226	11.5877	4.2559	[16]
			4.567	11.561	4.253	This work
Ga_3La_5	tP32(Ba_5Si_3)	P4/ncc	8.068	–	14.756	[17]
			8.09	–	14.764	This work
$GaLa_3$	cP4(AuCu3)	Pm $\overline{3}$ m	5.61	–	–	[18]
			4.94	–	–	This work
Ga_6Pr	tI14 (Ga_6Pu)	P4/nbm	6.014	–	7.649	[28]
			6.010	–	7.650	This work
Ga_2Pr	hP3(AlB_2)	P6/mmm	4.281	–	4.289	[28]
			4.291	–	4.293	This work
GaPr	oC8(CrB)	Cmcm	4.447	11.337	4.189	[27]
			4.447	11.336	4.189	This work
Ga_3Pr_5	tP32(Ba_5Si_3)	P4/ncc	7.939	–	14.50	[29]
			7.939	–	14.50	This work
$GaPr_2$	oP12(Co_2Si)	Pnma	6.690	5.188	9.714	[27]
			6.690	5.188	9.713	This work
Ga_4Pr_9	tI26(Ga_4Sm_9)	I4/m	12.008	–	5.182	[29]
			12.008	–	5.182	This work

2.2. Ga–Pr system

The phase diagram of the Ga–Pr system was studied by Iandelli [26] and Yatsenko et al. [6] using the differential thermal analysis (DTA) method. Four intermetallics were reported: Ga_2Pr, GaPr, Ga_3Pr_5 and Ga Pr_3. The compound Ga_2Pr melts congruently at 1470 °C, while a peritectic formation was reported for the intermetallic compounds GaPr, Ga_5Pr_3, and Ga Pr_3, at 1040, 865 and 720 °C respectively. The eutectic temperature on the Pr-rich side is at 868 °C, the eutectic on the Ga-rich side is degenerate and its temperature practically coincides with the melting temperature of pure Ga.

Cirafici [27] investigated the phase diagram in the range 0–50 at.% Ga using differential thermal, roentgengraphic and micrographic analysis. Four intermetallic compounds are reported: Pr_2Ga (peritectic formation at 705 °C), Pr_5Ga_3 (peritectic formation at 855 °C), GaPr (peritectic formation at 1015 °C) and Ga_2Pr (Eutectic formation at 1470 °C). An eutectic point occurs at about 18.5 at.% Ga and 580 °C. On the basis of previous works of [5,26] the presence of the $GaPr_3$ phase was to be excluded despite several attempts to obtain a homogeneous phase at this composition.

Later, by X-ray diffraction, differential thermal analysis, metallography, scanning electron microscopy and diffusion couples, Dayan et al. [28] reinvestigated the phase diagram in the range 66.7–100 at%. Ga and confirmed the existence of two new compounds: $PrGa_2$ and Ga_6Pr. The existence of the Ga_4Pr intermetallic is postulated on the basis of indirect experimental evidence. Ga_2Pr is the stoichiometric composition of the ϵ phase which forms congruently at 1470 °C [28] and has hexagonal structure of the AlB_2 type. Two compounds Pr_3Ga_2 [27] and Ga_4Pr_9 [29] were observed, but never confirmed.

Crystal structure and lattice parameters of the pure elements and six compounds (Ga_6Pr, Ga_2Pr, GaPr, Ga_3Pr_5, Ga_4Pr_9 and $GaPr_2$) are listed in Table 1 [27–29]. Ga_4Pr_9 were examined by Yatsenko et al. [29]. Dayan [28] has determined the structure of Ga_6Pr and Ga_2Pr. Ga_3Pr_5, GaPr and $GaPr_2$ were examined by Cirafici [27].

We have not found any thermodynamic data relating to this system, except the standard enthalpy of formation of the compound Ga_2Pr determined by Meschel and Kleppa [30] using high-temperature synthesis calorimetry and the enthalpies of formation of the binary compounds calculated by ab-initio method and given in the online databases: Materials project [31], OQMD [32], Aflow [33] and Jarvis [34].

3. Ab-initio calculations

In order to enhance the reliability of thermodynamic assessment, ab-initio calculations based on density functional theory (DFT) [35] as

implemented in the Vienna Ab-initio Simulation Package (VASP) [36, 37] with plane wave basis sets are performed. The calculations are performed using the Project Augmented Wave (PAW) pseudo-potential [38,39] and employed the Generalized Gradient Approximation (GGA) of Perdew, Burke and Ernzerhof (PBE) [40]. A plane wave cutoff energy of 500 eV is used for all compounds and K-points meshes are compiled using the fully automatic scheme [41]. Convergence on the number of K-points in the irreducible wedge of the first Brillouin zone is checked for each structure.

The energy of formation at 0 K of the Ga_pX_q compounds (X = La, Pr) is calculated by the following formula:

$$\Delta E\left(Ga_pX_q\right) = E\left(Ga_pX_q\right) - \frac{p}{p+q}E(Ga) - \frac{q}{p+q}E(X) \tag{1}$$

where E(Ga_pX_q), E (Ga) and E (X) are the total energies for the Ga_pX_q compound and pure elements in their stable structure. At 0 K, the energy of formation corresponds also to the enthalpy of formation.

4. Thermodynamic modeling

4.1. Pure elements

The Gibbs energy of the pure element i (i = Ga, La or Pr) in the phase φ referred to the enthalpy of its stable state at 298.15K is described as a function of temperature using the following equation:

$$\begin{aligned} G_i^{\varphi}(T) &= {}^0G_i^{\varphi}(T) - H^{SER}(298.15K) \\ &= a + bT + cTLnT + dT^2 + eT^3 + f^{-1} + gT^7 + hT^{-9} \end{aligned} \tag{2}$$

where H^{SER} (298.15 K) is the molar enthalpy of the element i at 298.15K in its standard element reference (SER) state. ${}^0G_i^{\varphi}$ and G_i^{φ} are the absolute and the relative Gibbs energy of the element i in the φ state, and T is the absolute temperature. In the present modeling, the Gibbs energies for pure elements were taken from the SGTE compilation of Dinsdale [42].

4.2. Intermetallic compounds

The Gibbs energy of the Ga_pX_q (X = La, Pr) compounds are given by the following expression:

$$\Delta G_f^{Ga_pX_q} = \Delta H_f^{Ga_pX_q} - T\Delta S_f^{Ga_pX_q} + \frac{p}{p+q}G_{Ga} + \frac{p}{p+q}G_X \tag{3}$$

The enthalpy and entropy of formation ($\Delta H_f^{Ga_pX_q}$ and $\Delta S_f^{Ga_pX_q}$) of the Ga_pX_q phase were considered as temperature independent optimizing parameters.

4.3. Solution phase

4.3.1. Substitutional solution model

The substitutional model was used to describe the primary solid solution phase αPr and ßPr in the Ga–Pr system. The Gibbs energy of phase φ is the sum of three terms: reference, ideal and excess term:

$$\begin{aligned} G^{\varphi} &= G_{ref}^{\varphi} + G_{ide}^{\varphi} + G_{exc}^{\varphi} \\ &= x_{Ga}G_{Ga}^{\varphi}(T) + x_{Pr}G_{Pr}^{\varphi}(T) + RT\left[x_{Ga}\,ln(x_{Ga}) + x_{Pr}ln(x_{Pr})\right] + G_{exc}^{\varphi} \end{aligned} \tag{4}$$

where R is the gas is constant, x_i is the mole fraction of specie i in the φ phase and T is the temperature. The excess Gibbs energy is described using the Redlich-Kister polynomial model [43] and given by the following expression:

$$G_{exc}^{\varphi} = x_{Ga}x_{Pr}\sum_{\vartheta=0}^{n} {}^{\vartheta}L_{Ga,Pr}^{\varphi}(x_{Ga} - x_{Pr})^{\vartheta} \tag{5}$$

where ${}^{\vartheta}L_{Ga,Pr}^{\varphi}$ (ϑ = 0, 1, 2…) are the interaction parameters which have been taken as temperature dependent and can be written as ${}^{\vartheta}L_{Ga,Pr}^{\varphi} = {}^{\vartheta}a_{Ga,Pr}^{\varphi} + {}^{\vartheta}b_{Ga,Pr}^{\varphi}T$. ${}^{\vartheta}a_{Ga,Pr}^{\varphi}$ and ${}^{\vartheta}b_{Ga,Pr}^{\varphi}$ are the parameters to be optimized.

4.3.2. Associate model

The liquid phase in the two systems presents very strong attractive interactions, and it is likely that the system will exhibit short-range ordering. The associate model [44] was chosen for the liquid phase in the present work with the constituent species Ga, Ga_2X and X (X = La, Pr). The Gibbs energy of the liquid is written as:

$$G^{Liq} = \sum_{i=Ga,X,Ga_2X} x_iG_i^{Liq} + RT\sum_{i=Ga,X,Ga_2X} x_i ln(x_i) + G_{exc}^{Liq} \tag{6}$$

where x_i shows the mole fraction of each species i (i = Ga, Ga_2X and X) in the liquid phase. G_{exc}^{Liq} is the excess Gibbs energy and is expressed as follows:

$$G_{exc}^{Liq} = \sum_i\sum_{i\neq j} x_ix_j\sum_{\vartheta=0}^{n} {}^{\vartheta}L_{i,j}^{Liq}(x_i - x_j)^{\vartheta} \tag{7}$$

The binary interaction parameters are described as linear functions of temperature ${}^{\vartheta}L_{i,j}^{Liq} = {}^{\vartheta}a_{i,j}^{Liq} + {}^{\vartheta}b_{i,j}^{Liq}T$.

4.3.3. Sublattice model

The Ga_2La phase is described with two sublattice model $(Ga,La)_2(Ga,La)_1$.The three terms of the Gibbs energy function are:

$$G_{ref}^{Ga_2La}(T) = y_{Ga}^{'}y_{Ga}^{''}G_{Ga:Ga}^{Ga_2La} + y_{Ga}^{'}y_{La}^{''}G_{Ga:La}^{Ga_2La} + y_{La}^{'}y_{Ga}^{''}G_{La:Ga}^{Ga_2La} + y_{La}^{'}y_{La}^{''}G_{La:La}^{Ga_2La} \tag{8}$$

$$G_{ide}^{Ga_2La} = RT\left[\left(y_{Ga}^{'}\ln\left(y_{Ga}^{'}\right) + y_{La}^{'}ln\left(y_{La}^{'}\right)\right) + 2\left(y_{Ga}^{''}ln\left(y_{Ga}^{''}\right) + y_{La}^{''}ln\left(y_{La}^{''}\right)\right)\right] \tag{9}$$

$$\begin{aligned} G_{exc}^{Ga_2La} &= y_{Ga}^{'}y_{La}^{'}y_{Ga}^{''}\sum_{\vartheta=0}^{n} {}^{\vartheta}L_{Ga,La:Ga}^{Ga_2La}\left(y_{Ga}^{'} - y_{La}^{'}\right)^{\vartheta} + y_{Ga}^{'}y_{La}^{'}y_{La}^{''} \\ &\times \sum_{\vartheta=0}^{n} {}^{\vartheta}L_{Ga,La:La}^{Ga_2La}\left(y_{Ga}^{'} - y_{La}^{'}\right)^{\vartheta} + y_{Ga}^{'}y_{Ga}^{''}y_{La}^{''}\sum_{\vartheta=0}^{n} {}^{\vartheta}L_{Ga:Ga,La}^{Ga_2La}\left(y_{Ga}^{''} - y_{La}^{''}\right)^{\vartheta} \\ &+ y_{La}^{'}y_{Ga}^{''}y_{La}^{''}\sum_{\vartheta=0}^{n} {}^{\vartheta}L_{La:Ga,La}^{Ga_2La}\left(y_{La}^{''} - y_{Ga}^{''}\right)^{\vartheta} \end{aligned} \tag{10}$$

where $y_{Ga}^{'}$ and $y_{La}^{'}$ are the site fraction of Ga and La in the first sublattice, $y_{Ga}^{''}$ and $y_{La}^{''}$ the site fraction of Ga and La in the second sublattice. $G_{Ga:Ga}^{Ga_2La}$, $G_{Ga:La}^{Ga_2La}$ and $G_{La:La}^{Ga_2La}$ are the Gibbs energy of the hypothetical compounds $GaGa_2$, $GaLa_2$ and $LaLa_2$, respectively. $G_{La:Ga}^{Ga_2La}$ is the formation Gibbs energy of the stoichiometric intermetallic Ga_2La. The enthalpies of formation of the Ga_2La compound and the hypothetical compounds $GaGa_2$, $GaLa_2$ and $LaLa_2$ will be calculated by using the VASP code.

4.4. Optimization methodology

Experimental data cited above and calculated enthalpies of formation of the binary compounds were used in the thermodynamic evaluation of all phases in the Ga–La and Ga–Pr systems. The thermodynamic optimization of the Gibbs energy parameters is an application of the CALPHAD approach [45,46]. It was carried out by using the PARROT module in the ThemoCalc software [47]. A set of optimized model parameters are obtained, each experimental datum is weighted carefully according to its merits, and changed by trial and error during the assessment, reaching a good agreement with the experimental data within the expected uncertainty limits.

5. Results and discussion

5.1. Ab-initio calculations

5.1.1. The known compounds

First we have studied the stable and known structure compounds in the two systems. Lattice parameters and enthalpies of formation are reported in Tables 1–3. The calculated lattice parameters are in good agreement with experimental data, with a maximum difference about 2.62%.

5.1.2. Ga_4Pr compound

The existence of the intermetallic Ga_4Pr has been reported by Dayan et al. [28] but without structural properties. However, it was never confirmed in any other work. To calculate the enthalpy of formation of this supposed compound, we have proposed 17 crystal structures with A_4B stoichiometric. The calculated lattice parameters and their enthalpies of formation are summarized in Table 4.

The lower enthalpy is obtained with oP20 structure (space group Pmmm) but above the line of the ground state by about 7 kJ/mol. So it may not be stable at 0 K.

Table 2
The standard enthalpies of formation of the intermetallic compounds in the Ga–La system.

Compound	ΔH (kJ/mol)	Method	Ref.
Ga_6La	−35.99	Ab-initio	This work
	−35.82	Calphad	This work
	−41.94	Calphad	[25]
	−37.5 ± 3.7	Calorimetry	[22]
	−42.70	Miedema's model	[23]
	−35.17	Ab-initio[a]	[31]
	−35.75	Ab-initio[a]	[32]
Ga_4La	−44.30	Ab-initio	This work
	−45.69	Calphad	This work
	- 55.03	Calphad	[25]
	−49.34	Ab-initio[a]	[33]
Ga_2La	−67.91	Ab-initio	This work
	−67.39	Calphad	This work
	−72.88	Calphad	[25]
	−69 ± 3.7	Calorimetry	[22]
	−69.2 ± 2.4	Calorimetry	[21]
	−74.60	EMF	[20]
	−67.50	Miedema's model	[23]
	−67.84	Ab-initio[a]	[31]
	−67.55	Ab-initio[a]	[32]
	−69.67	Ab-initio[a]	[33]
GaLa	−57.48	Ab-initio	This work
	−57.77	Calphad	This work
	- 77.74	Calphad	[25]
	−74.8 ± 3.7	calorimetry	[22]
	−73.50	Miedema's model	[23]
	−57.05	Ab-initio[a]	[31]
	−56.86	Ab-initio[a]	[32]
	−59.17	Ab-initio[a]	[33]
	−59.12	Ab-initio[a]	[34]
Ga_3La_5	−44.18	Ab-initio	This work
	−43.62	Calphad	This work
	−58.51	Calphad	[25]
	−62.6 ± 4.5	calorimetry	[22]
	−61.80	Miedema's model	[23]
	−44.52	Ab-initio[a]	[31]
	−45.20	Ab-initio[a]	[32]
	−42.11	Ab-initio[a]	[33]
$GaLa_3$	−25.23	Ab-initio	This work
	−29.49	Calphad	This work
	−38.63	Calphad	[25]
	−41.00	Miedema's model	[23]
	−26.69	Ab-initio[a]	[32]
	−28.71	Ab-initio[a]	[33]
	−26.92	Ab-initio[a]	[34]

[a] Data obtained from the online databases: Materials project [31], OQMD [32], Aflow [33] and Jarvis [34].

Table 3
The standard enthalpies of formation of the intermetallic compounds in the Ga–Pr system.

Compound	ΔH (kJ/mol)	Method	Ref.
Ga_6Pr	−34.65	Ab-initio	This work
	−34. 96	Calphad	This work
	−33.34	Ab-initio[a]	[31]
	−34.21	Ab-initio[a]	[32]
Ga_4Pr	−40.81	Ab-initio	This work
Ga_3Pr	−56,59	Ab-initio	This work
	−56.70	Calphad	This work
	−55.12	Ab-initio[a]	[31]
	−55.60	Ab-initio[a]	[32]
	−55.28	Ab-initio[a]	[34]
Ga_2Pr	−67.49	Ab-initio	This work
	−67.68	Calphad	This work
	−69.3 ± 2.7	Calorimetry	[30]
	−66.88	Ab-initio[a]	[31]
	−66.69	Ab-initio[a]	[32]
	−67.28	Ab-initio[a]	[34]
GaPr	−57.41	Ab-initio	This work
	−57.78	Calphad	This work
	−55.79	Ab-initio[a]	[31]
	−55.99	Ab-initio[a]	[32]
	−57.01	Ab-initio[a]	[34]
Ga_3Pr_5	−44.98	Ab-initio	This work
	−44.97	Calphad	This work
	−42.11	Ab-initio[a]	[31]
	−43.08	Ab-initio[a]	[32]
$GaPr_2$	−38.58	Ab-initio	This work
	−39.99	Calphad	This work
	−36.71	Ab-initio[a]	[32]

[a] Data obtained from the online databases: Materials project [31], OQMD [32], Aflow [33] and Jarvis [34].

Table 4
Enthalpies of formation of Ga_4Pr in different suspected structures.

Prototype	Pearson symbol	Space Group Number	Lattice parameters (°A)			ΔH (kJ/mol)
			a	b	c	
PtSn4	oC20	41	6.388	6.419	11.357	−9,16522
B4Mg	oP20	62	5.464	4.428	7.472	−4,21814
PdSn4	oC20	68	6.442	11.445	6.389	314,62891
MoNi4	tI10	87	5.720		3.564	−16,55231
Cl4Th	tI20	88	6.408		12.924	34,09932
NbTe4	tP10	103	6.551		6.847	52,76802
Pd4Se	tP10	114	5.232		5.647	−7,43335
NbTe4	tP10	124	6.499		6.837	50,59862
Pb4Pt	tP10	125	6.640		5.970	46,48184
B4Th	tP20	127	7.256		4.113	22,42079
Al4Ba	tI10	139	4.530		11.100	−40,71718
Ge4Ir	hP15	152	6.215		7.784	26,69541
Ga4Ni	cI40	197	8.430			−12,13146
CFe4	cP5	215	3.878			13,98320
F4Si	cI10	217	5.459			36,52282
Hg4Pt	cI10	229	6.186			36,79570
Ga_4La	oP20	47	4.355	12.956	7.745	−40,81075

To confirm the existence of this compound, we calculated its elastic stiffness constants. The stability condition for the orthorhombic structures given by Wu et al. [48]:

$$C_{11} > 0,\ C_{22} > 0,\quad C_{33} > 0,\quad C_{44} > 0,\quad C_{55} > 0,\quad C_{66} > 0$$

$$[C_{11} + C_{22} + C_{33} + 2(C_{12} + C_{13} + C_{23})] > 0$$

$$(C_{11} + C_{22} - 2C_{12}) > 0$$

$$(C_{11} + C_{33} - 2C_{13}) > 0$$

$$(C_{22} + C_{33} - 2C_{23}) > 0$$

The calculated elastic stiffness constants of the oP20 structure are

listed in Table 5. One can notice that the two last conditions are not satisfied, so the Ga_4Pr compound is mechanically instable and will not be taken into account in the Calphad optimization.

5.1.3. Suspected compound: Ga_3Pr

The high similitude between intermetallics in the range 0–50 at% in the Ga-RE systems (RE = Er, Ho, Lu, Tm, Sc, Tb, Pr), pushed us to suspect the existence of the Ga_3Pr compound according to the existence of Ga_3Er, Ga_3Ho, Ga_3Lu, Ga_3Tm, Ga_3Sc and Ga_3Tb. So we chose for this compound the structure Hexagonal Ni_3Sn prototype, and calculated its enthalpy of formation. The calculated lattice parameters and their enthalpies of formation are given in Table 6.

We have also determined the elastic stiffness constants to confirm the mechanical stability of this compound (Table 7). These values satisfy all stability conditions for Hexagonal structures given by Wu et al. [48]:

$$C_{12} > 0, C_{33} > 0, C_{44} > 0, C_{66} > 0,$$

$$(C_{11} + C_{12})(C_{33} * C_{33} - 2\, C_{13} * C_{13}) > 0$$

5.1.4. Hypothetical compounds in the Ga_2La phase

The Gibbs energy of the Ga_2La phase is described by the sublattice model with three hypothetical compounds $GaGa_2$, $GaLa_2$ and $LaLa_2$. So in order to get a reliable optimization we have calculated the enthalpies of formation of these compounds in the Ga_2La structure (Table 8). The calculated enthalpies of formation will be introduced as initial values in the thermodynamic optimization of the Ga–La system by the Calphad method.

5.2. Thermodynamic assessment

5.2.1. Ga–La system

The parameters of the thermodynamic description of the Ga–La system obtained in the present assessment are given in Table 9. The calculated Ga–La phase diagram is shown in Fig. 1 and compared with the experimental data of Yatsenko et al. [6], Lu et al. [8], Pelleg et al. [7], Kaplan and Kimmel [12,13] and assessed by Idbenali and Servant [25] in Fig. 2. The calculated liquidus curve is more close to the experimental data [6,8] than one assessed previously by Idbenali and Servant [25]. As we can see in Fig. 2 a small difference in the liquidus is observed for the composition range 60–75 at.% Ga, which is a little higher than the experimental data of Yatsenko et al. [6]. The comparison of the calculated invariant reactions with the experimental and assessed ones in this system is listed in Table 10.

Fig. 3 shows the enthalpies of formation of the Ga–La intermetallic compounds determined by Calphad method and ab-initio calculations together with the experimental data obtained by Meshel and Kleppa [21], Babu et al. [22], Vnuchkova et al. [20], estimated by the Midema's model [23], as well as the calculated values determined by ab-initio method from Wang et al. [24] and given in the online databases (Materials project [31], OQMD [32], Aflow [33] and Jarvis [34]). The calculated and optimized enthalpies of formation are in good agreement with those calculated by ab-initio method and measured by calorimetry and they are acceptable in comparison with the predicted ones [23], especially if we take into account the large uncertainties of the Miedema model.

5.2.2. Ga–Pr system

The different experimental studies of the Ga–Pr system [6,27,28] confirmed the existence of five intermetallics Ga_6Pr, Ga_2Pr, GaPr, Ga_3Pr_5 and $GaPr_2$. The discrepancy is in the existence of the compound Ga_3Pr or Ga_4Pr. According to studies carried out by Dayan et al. [9] and Pelleg et al. [49] the heavy lanthanides form RGa_3 compound and the RGa_2 is a stoichiometric phase, but the light lanthanides are unable to form RGa_3 compound and the RGa_2 is given with homogeneity range. All studies carried out on the Ga–Pr system have not confirmed the homogeneity range of the Ga_2Pr phase as well as the existence of the Ga_3Pr and Ga_4Pr compounds. In order to clarified the existence of these two compounds we have calculated the enthalpies of formation of the Ga_3Pr

Table 6
Enthalpy of formation and lattice parameters of Ga_3Pr compound.

Prototype	Pearson symbol	Space group	Lattice parameters (°A)			ΔH (kJ/mol)
			a	b	c	
Ni_3Sn	hP8	P63/mmc	6.5422	5.6657	4.5838	−56.59

Table 7
Elastic stiffness constants of the Ga_3Pr compound.

C_{11}	C_{12}	C_{13}	C_{33}	C_{44}	C_{66}
132.1	32.1	37.3	127.6	57.4	50.0

Table 8
Enthalpies of formation of the hypothetical compounds in the Ga_2La phase.

	$GaGa_2$	$LaLa_2$	$GaLa_2$
ΔH (kJ/mol)	18.25	3.47	−22.10

Table 9
Thermodynamic parameters of the Ga–La system.

Phase	Parameters
Liquid	$G^{Liq}_{Ga_2La} = -87348 + 80.82T + 2G^{Liq}_{Ga} + G^{Liq}_{La}$
	${}^0L^{Liq}_{Ga,La} = -171800 + 23.51T$
	${}^1L^{Liq}_{Ga,La} = -24206 - 9.05T$
	${}^2L^{Liq}_{Ga,La} = +12400$
	${}^0L^{Liq}_{La,\ Ga_2La} = -90247 + 63.37T$
	${}^0L^{Liq}_{Ga,\ Ga_2La} = -100117 + 92.18T$
Ga_2La	$G^{Ga_2La}_{Ga:Ga} = 62910 + 3GHSERGA$
	$G^{Ga_2La}_{La:La} = 10887 + 3GHSERLA$
	$G^{Ga_2La}_{Ga:La} = -202176 + 11.74T + 2GHSERGA + GHSERLA$
	$G^{Ga_2La}_{La:Ga} = -61044 + GHSERGA + 2GHSERLA$
	${}^0L^{Ga_2La}_{Ga:Ga,La} = -99246 + 20.44T$
	${}^1L^{Ga_2La}_{Ga:Ga,La} = -8376 + 15.00T$
	${}^2L^{Ga_2La}_{Ga:Ga,La} = -78246$
	${}^0L^{Ga_2La}_{La:Ga,La} = -45246 + 4.20T$
	${}^0L^{Ga_2La}_{Ga,La:La} = +75000 + 14.40T$
Ga_6La	$\Delta G^{Ga_6La} = -35821 + 3.09T$
Ga_4La	$\Delta G^{Ga_4La} = -45690 + 3.047T$
GaLa	$\Delta G^{GaLa} = -57779 + 3.450T$
Ga_3La_5	$\Delta G^{Ga_3La_5} = -43622 - 0.10T$
$GaLa_3$	$\Delta G^{GaLa_3} = -29494 - 0.56T$

Table 5
Elastic stiffness constants of the Ga_4Pr compound.

C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
−747.7	−386.2	445.5	154.7	437.8	109.1	56.2	26.2	50.5

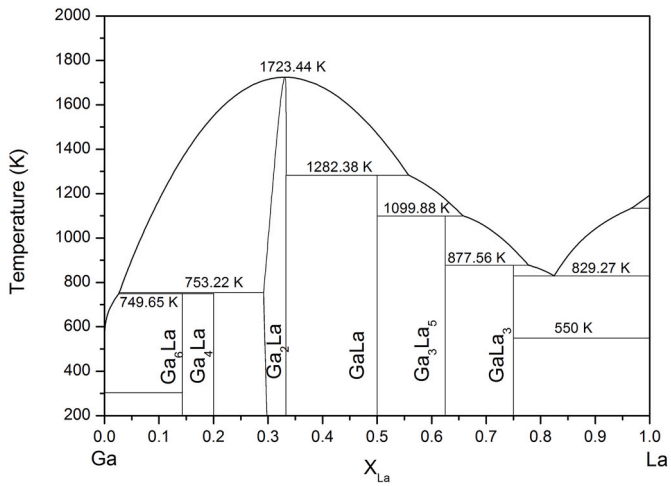

Fig. 1. Calculated Ga–La phase diagram.

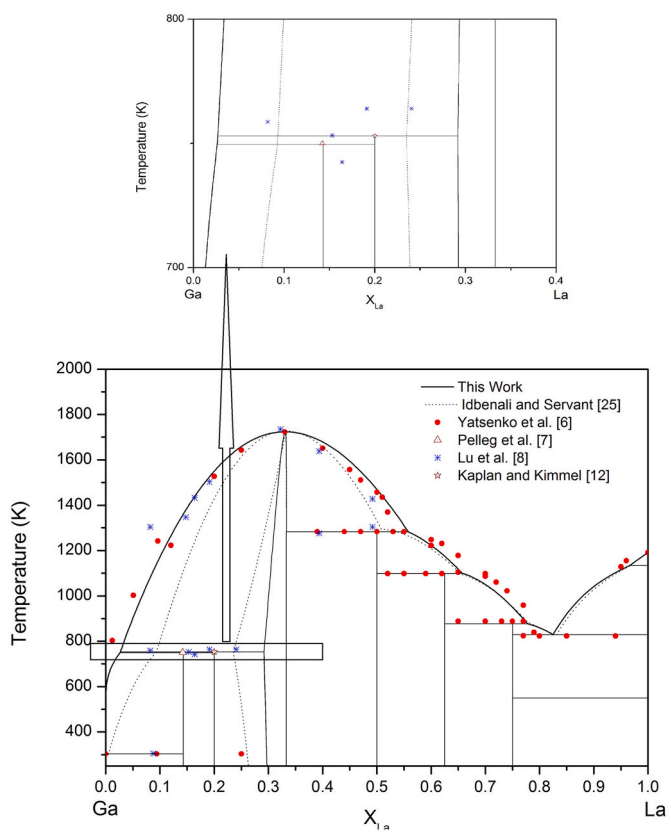

Fig. 2. Calculated Ga–La phase diagram compared with experimental and assessed data.

and Ga_4Pr compounds and determined their mechanical stabilities. The Ga_3Pr compound was found to be energetically and mechanically stable, unlike the Ga_4Pr compound. Therefore in the present assessment, we have taken into account the Ga_3Pr compound and the Ga_2Pr is treated as a stoichiometric compound.

We have used as data some equilibrium points of the experimental diagram [6,27,28] as well as the enthalpies of formation of the six compounds that have already been calculated by the ab-initio method. The calculated thermodynamic quantities of all phases are summarized in Table 11. The comparison between calculated and experimental Ga–Pr phase diagram is shown in Fig. 4. The calculated liquidus in

Table 10
Temperatures and compositions of the invariant reactions in the Ga–La system.

Reaction	Type of reaction	T (K)	X(Liq) at.% La	Reference
Liq↔Ga + Ga_6La	Eutectic	302.91	4.65 E−13	This work
		302	<0.1	[4,6]
		301.7	0.5	[25]
Liq + Ga_4La↔ Ga_6La	Peritectic	749.65	2.62	This work
		750	0.5	[7]
		748.6	9.2	[25]
Liq + Ga_2La ↔ Ga_4La	Peritectic	753.22	2.70	This work
		753	–	[8,12]
		752.3	9.3	[25]
Liq ↔ Ga_2La	Congruent	1723.44	33.33	This work
		1723	33.33	[4,6]
		1727	33.33	[25]
Liq + Ga_2La ↔ GaLa	Peritectic	1282.38	55.78	This work
		1283	55.55	[4,6]
		1295	50.8	[25]
Liq + GaLa ↔ Ga_3La_5	Peritectic	1099.88	65.74	This work
		1098	70	[4,6]
		1095	65.5	[25]
Liq + Ga_3La_5↔ $GaLa_3$	Peritectic	877.56	77.74	This work
		888	79	[4,6]
		885.8	77.7	[25]
Liq ↔ $GaLa_3$+ FCC	Eutectic	829.27	82.47	This work
		823	80	[4,6]
		826	83.0	[25]
Liq +βLa↔γLa	Catatectic	1134.00	96.62	This work
		1138	94	[4,6]
		1134	96.79	[25]

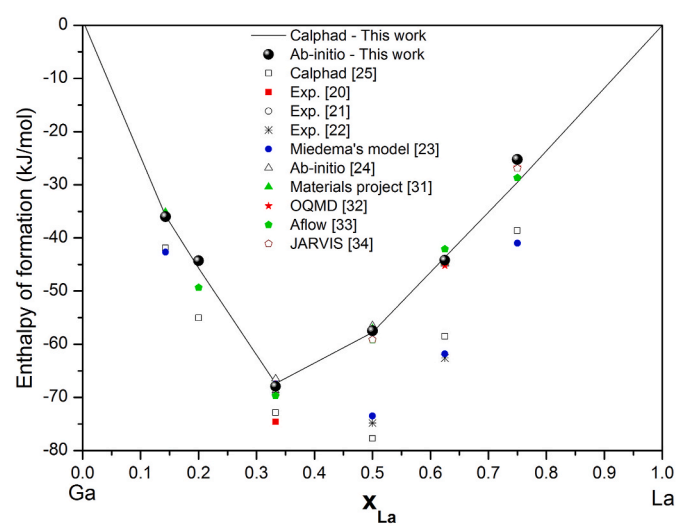

Fig. 3. Enthalpies of formation of the intermetallic compounds in the Ga–La system.

Ga-rich part is in good agreement with the experimental data of Dayan et al. [28], while in the Pr-rich part our result matches well with data of Iandelli [26] and Yatsenko et al. [6]. All the invariant reactions are well reproduced in comparison with experimental data (Table 12).

Fig. 5 shows the enthalpies of formations of the intermetallics in the Ga–Pr system. A comparison based on the results of the Calphad method and the ab-initio calculation shows a very high coherence and correspondence and a good agreement with the experimental [30] and the calculated data [31,32,34].

6. Conclusion

The Ga–La and Ga–Pr systems have been studied by Combination of ab-initio calculations and CALPHAD approach. The enthalpies of formation of the confirmed and suspect compounds in the two systems are

Table 11
Thermodynamic parameters of the Ga–Pr system.

Phase	Parameters
Liquid	$G^{Liq}_{Ga_2Pr} = -97348 + 96.82T + 2G^{Liq}_{Ga} + G^{Liq}_{Pr}$
	${}^{0}L^{Liq}_{Ga,Pr} = -171920 + 37.00T$
	${}^{1}L^{Liq}_{Ga,Pr} = -36095 - 9.808T$
	${}^{2}L^{Liq}_{Ga,Pr} = -5497$
	${}^{0}L^{Liq}_{Pr,\ Ga_2Pr} = -43247 + 58.37T$
	${}^{0}L^{Liq}_{Ga,\ Ga_2Pr} = -90117 + 82.18T$
BCC	${}^{0}L^{BCC}_{Ga,Pr} = -150988 + 44T$
	${}^{1}L^{BCC}_{Ga,Pr} = -30845 + 8T$
DHCP	${}^{0}L^{DHCP}_{Ga,Pr} = -129383 + 44T$
	${}^{1}L^{DHCP}_{Ga,Pr} = -12847 + 4T$
Ga_6Pr	$\Delta G^{Ga_6Pr} = -34968 + 2.268\ T$
Ga_3Pr	$\Delta G^{Ga_3Pr} = -56700 + 7.450\ T$
Ga_2Pr	$\Delta G^{Ga_2Pr} = -67680 + 6.280T$
GaPr	$\Delta G^{GaPr} = -57787 + 6.760T$
Ga_3Pr_5	$nG^{Ga_3Pr_5} = -44974 + 4.470T$
$GaPr_2$	$\Delta G^{GaPr_2} = -39999 + 3.672T$

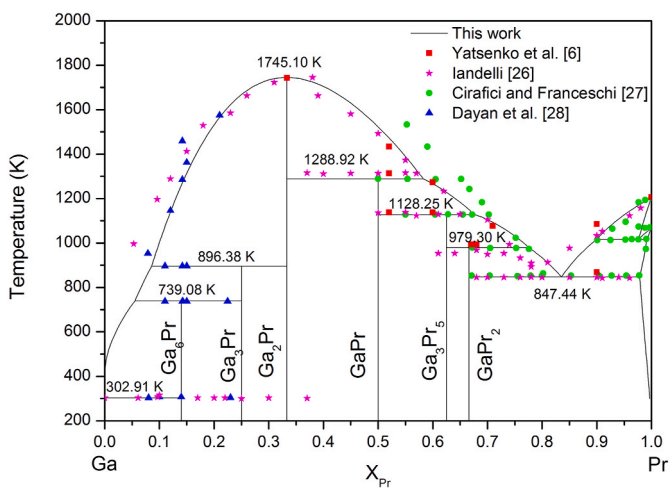

Fig. 4. Comparison between calculated and experimental phase diagram of the Ga–Pr system.

determined. The ground state of the two systems reveals that six compounds are stable in each system. In the Ga–Pr system the Ga_3Pr compound is stable instead of the Ga_4Pr. The Ga–La system has been reassessed and the Ga–Pr system has been assessed for the first time using the experimental phase diagrams data and enthalpies of formation of the intermetallic phases of the two systems obtained by ab-initio calculations. A good agreement between calculation and experiment was found. According to the results obtained in this work, we strongly suggest that the continuation of the experimental investigation is necessary for the Ga–La and Ga–Pr systems to confirm the inferences deduced in particular in the Ga-rich parts.

Data availability

All the thermodynamic parameters required to reproduce the results of the present study are available in supplementary materials.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 12
Temperatures and compositions of the invariant reactions in the Ga–Pr system.

Reaction	Type of reaction	Temperature (K)	x(Liq) at.% Pr	Ref.
Liq ↔ Ga + Ga_6Pr	Eutectic	302.91	1.3E-7	This work
		302 ± 2		[28]
Liq + Ga_3Pr↔ Ga_6Pr	Peritectic	739.08	5.52	This work
		739 ± 2		[28]
Liq + Ga_2Pr ↔ Ga_3Pr	Peritectic	896.38	8.57	This work
		897 ± 5		[28]
Liq ↔ Ga_2Pr	Congruent	1745.10	33.33	This work
		1743	33.33	[6]
Liq + Ga_2Pr ↔ GaPr	Peritectic	1288.92	58.25	This work
		1288		[27]
		1313		[6]
Liq + GaPr ↔ Ga_3Pr_5	Peritectic	1128.25	67.64	This work
		1128		[27]
		1138		[6]
Liq + Ga_3Pr_5↔ $GaPr_2$	Peritectic	979.30	77.58	This work
		978		[27]
		993		[6]
Liq↔ $GaPr_2$ + DHCP	Eutectic	847.44	83.52	This work
		853		[27]
		868		[6]
Liq + DHCP↔ BCC	Catatectic	1017.72	90.63	This work
		1013		[27]

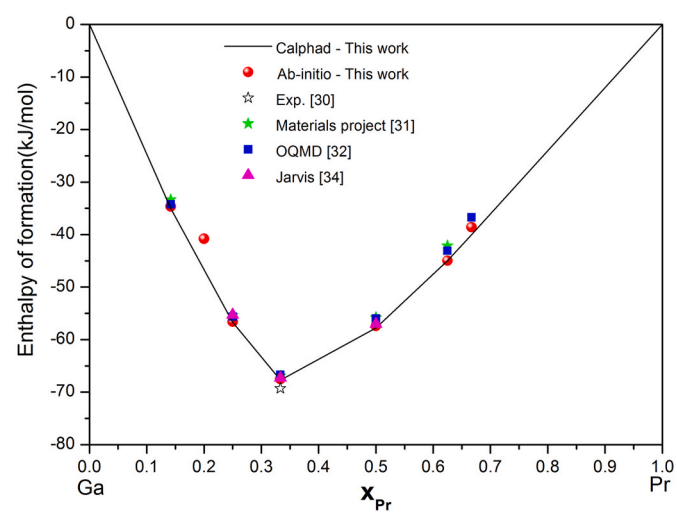

Fig. 5. Enthalpies of formation of the intermetallic compounds in the Ga–Pr system.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2021.102387.

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